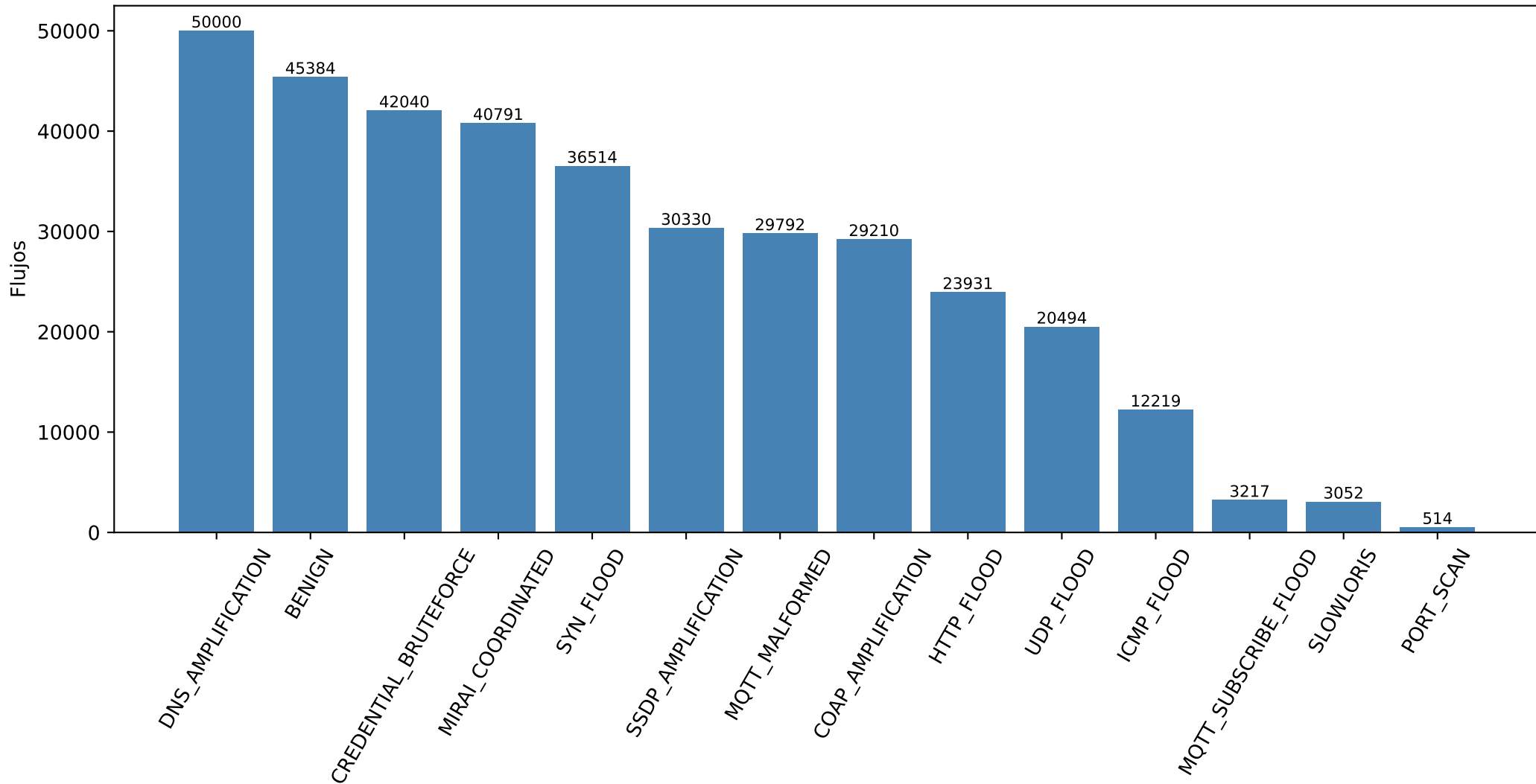


IoT-SDN-AI — Reporte de evaluación

run: run_20260924-214452

Dataset:	505,768 flujos
Features:	50
Clases:	14 (BENIGN, COAP_AMPLIFICATION, CREDENTIAL_BRUTEFORCE, DNS_AMPLIFICATION, HTTP_FLOOD...)
Modelos:	RandomForest, XGBoost, MLP, IsolationForest, AutoEncoder
Estrategia:	grouped split anti-fuga (StratifiedGroupKFold fold0) 80/20, StandardScaler

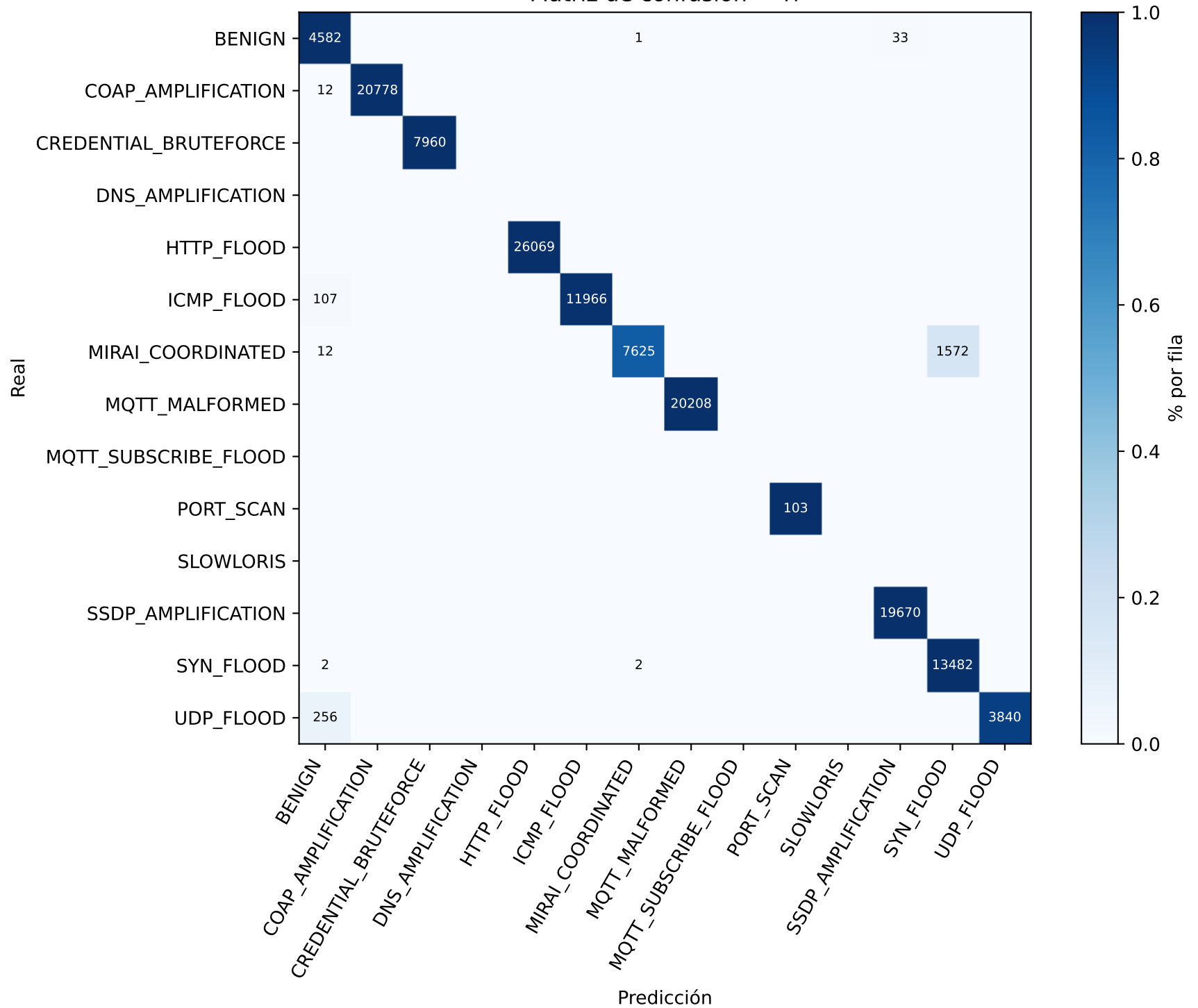
Distribución de clases en el dataset



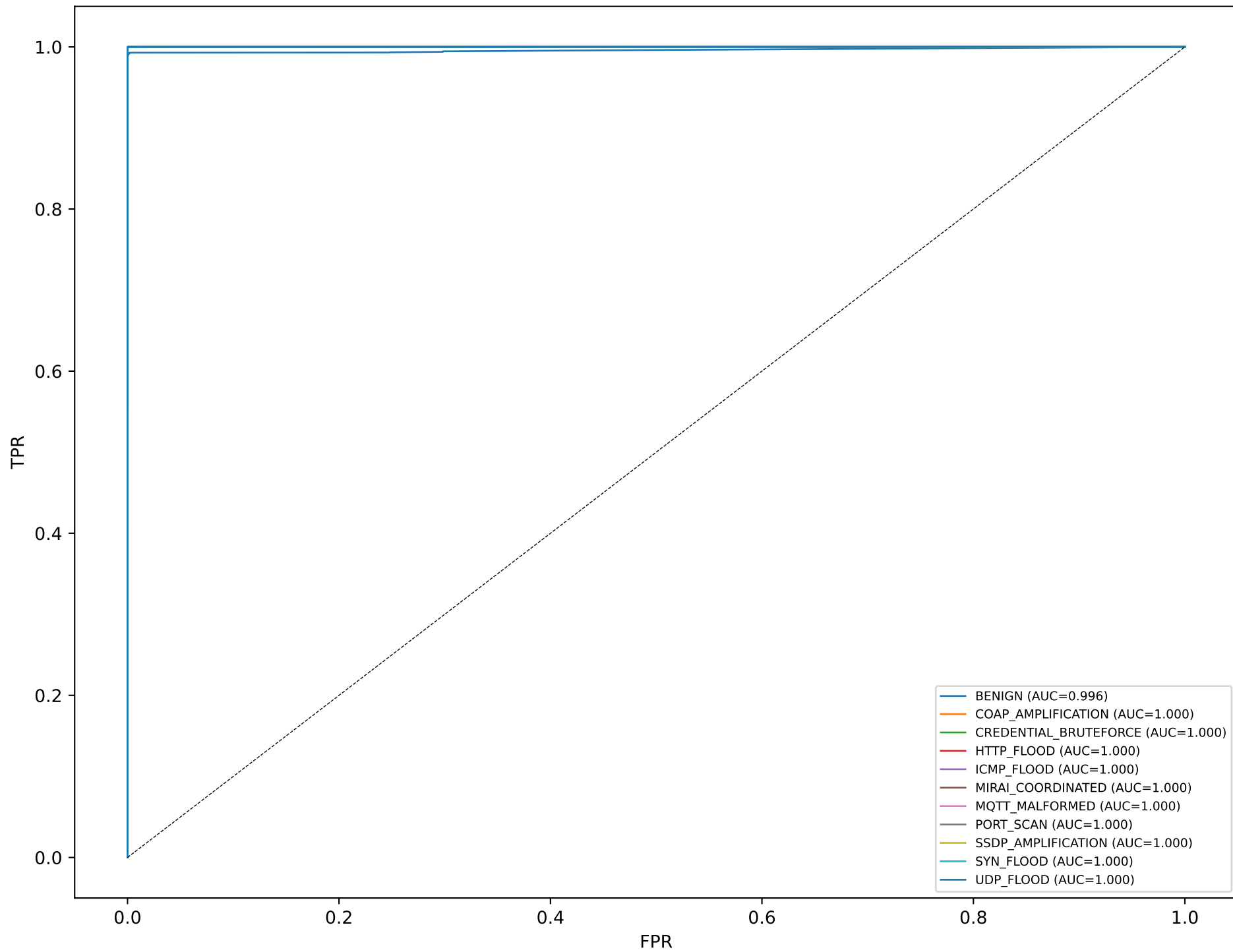
Classification report — rf

	precision	recall	f1-score	support
BENIGN	0.92	0.99	0.96	4616
COAP_AMPLIFICATION	1.00	1.00	1.00	20790
CREDENTIAL_BRUTEFORCE	1.00	1.00	1.00	7960
HTTP_FLOOD	1.00	1.00	1.00	26069
ICMP_FLOOD	1.00	0.99	1.00	12073
MIRAI_COORDINATED	1.00	0.83	0.91	9209
MQTT_MALFORMED	1.00	1.00	1.00	20208
PORT_SCAN	1.00	1.00	1.00	103
SSDP_AMPLIFICATION	1.00	1.00	1.00	19670
SYN_FLOOD	0.90	1.00	0.94	13486
UDP_FLOOD	1.00	0.94	0.97	4096
accuracy			0.99	138280
macro avg	0.98	0.98	0.98	138280
weighted avg	0.99	0.99	0.99	138280

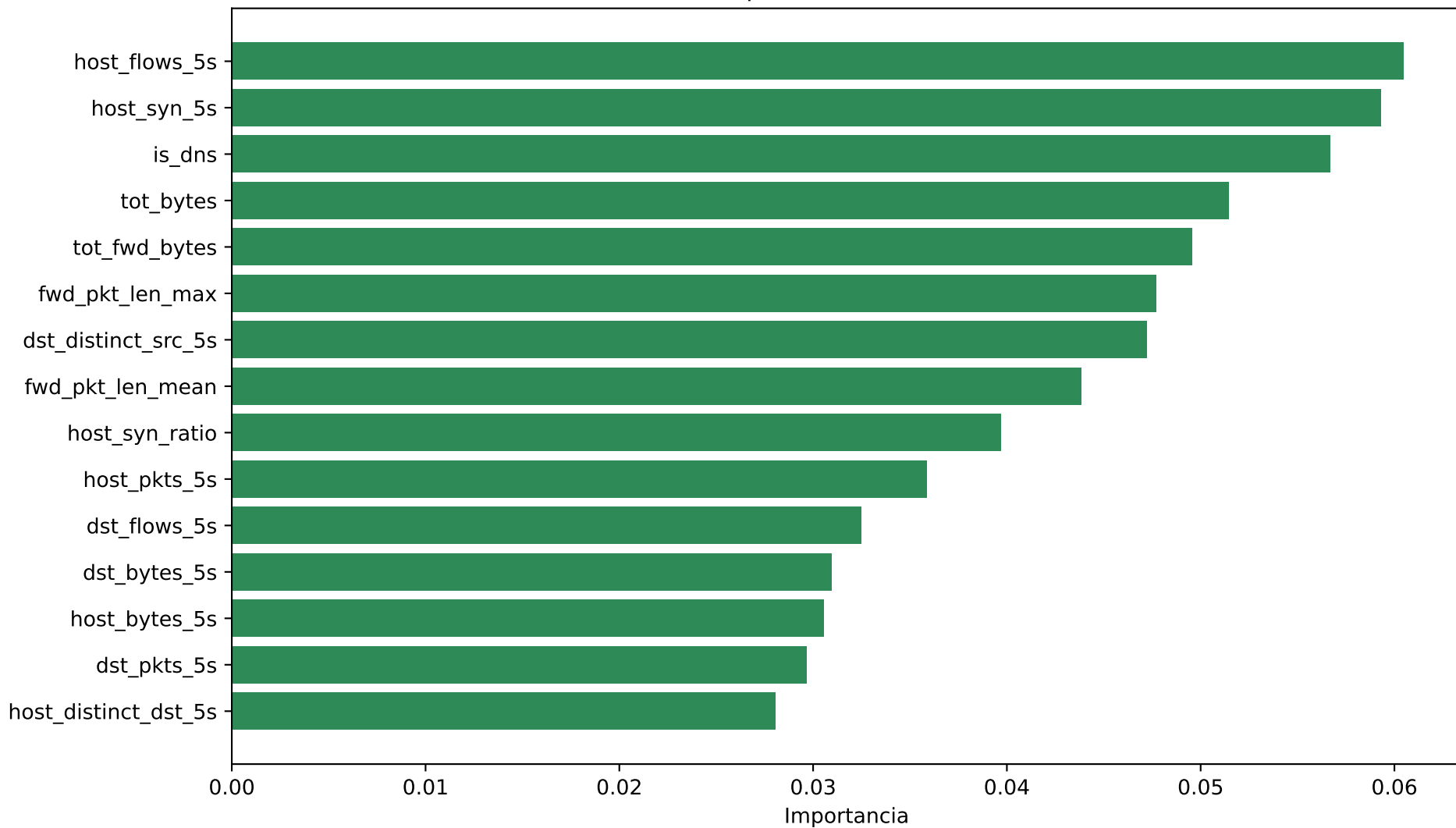
Matriz de confusión — rf



ROC OvR — rf



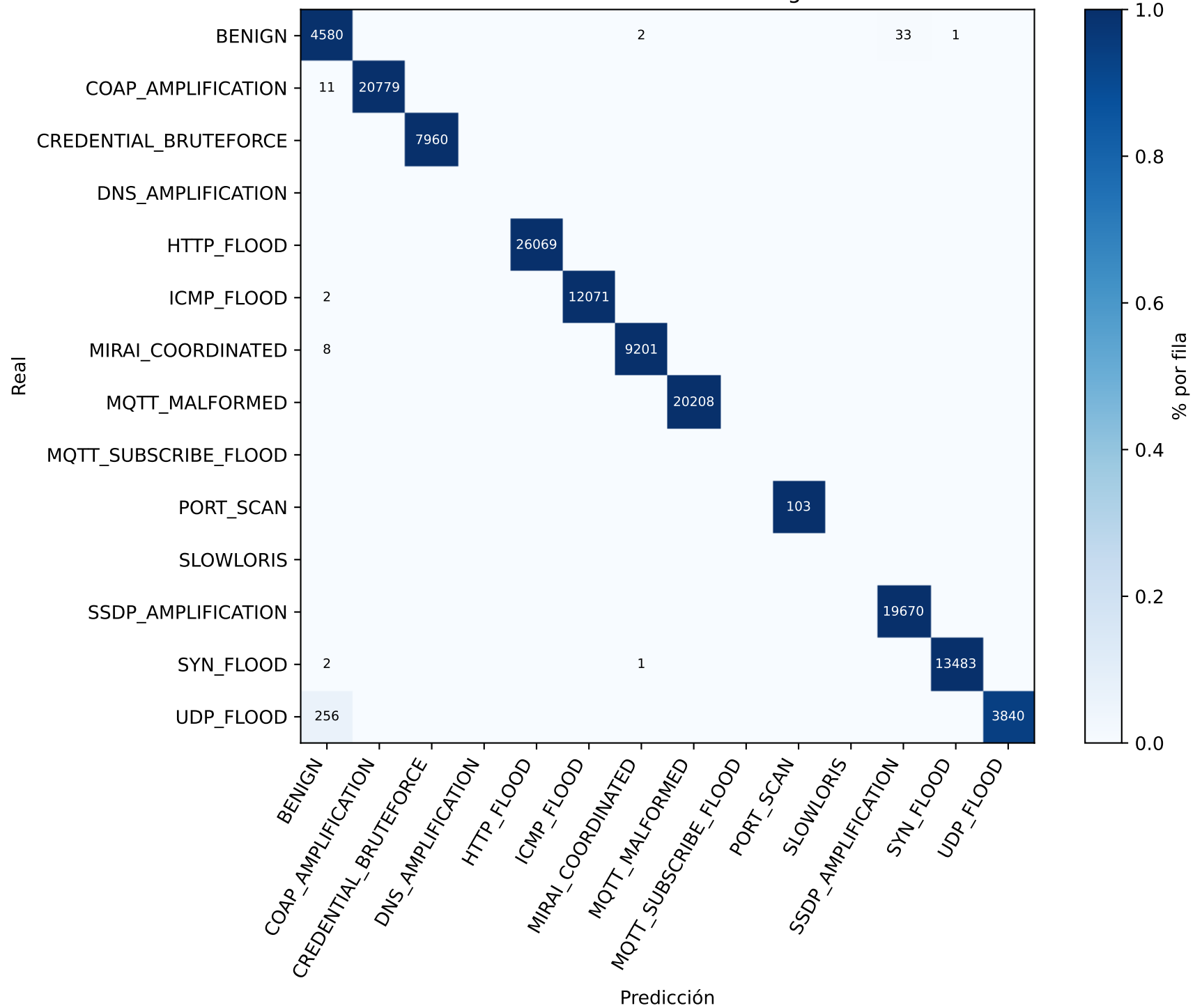
Top-15 features — rf



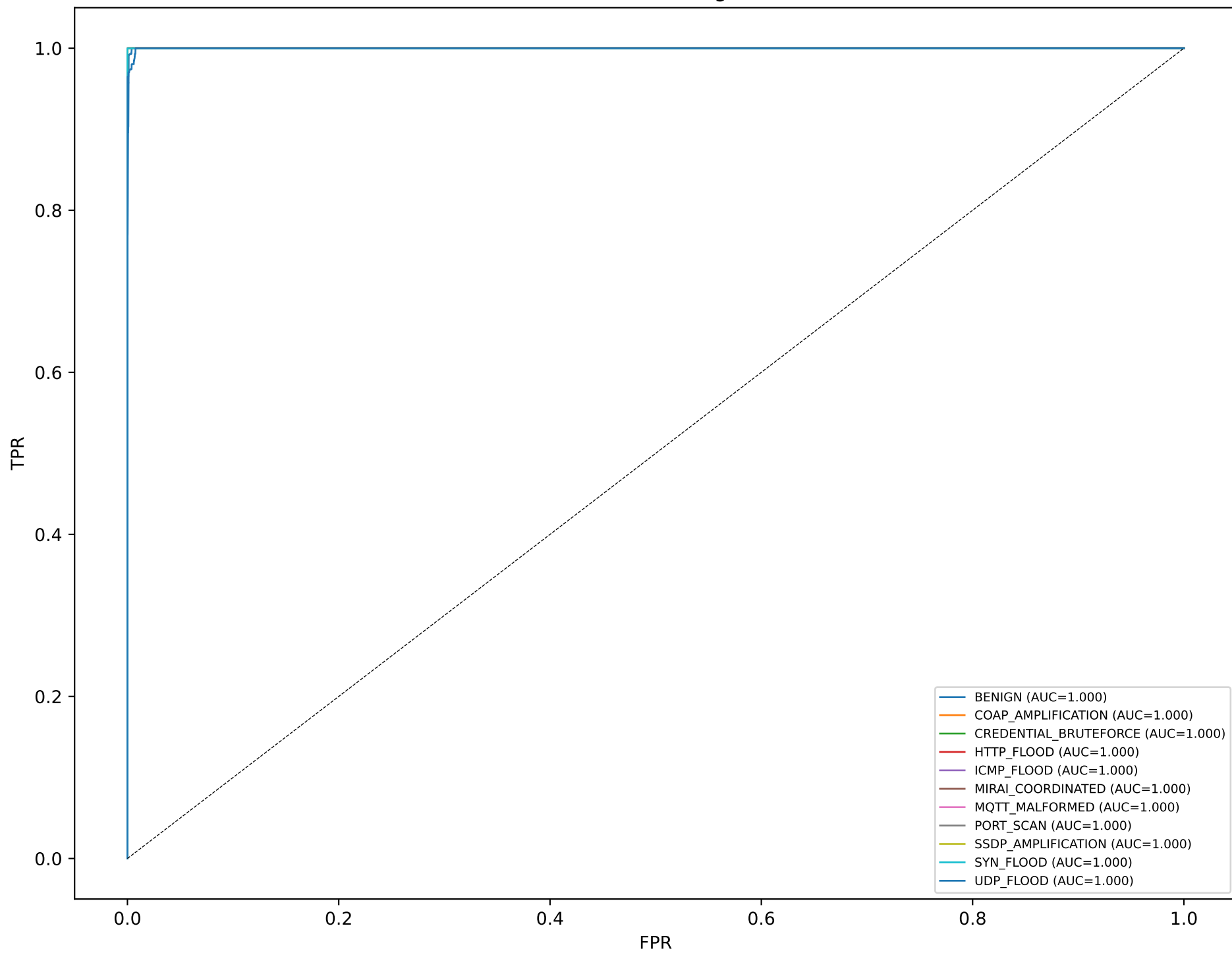
Classification report — xgb

	precision	recall	f1-score	support
BENIGN	0.94	0.99	0.97	4616
COAP_AMPLIFICATION	1.00	1.00	1.00	20790
CREDENTIAL_BRUTEFORCE	1.00	1.00	1.00	7960
HTTP_FLOOD	1.00	1.00	1.00	26069
ICMP_FLOOD	1.00	1.00	1.00	12073
MIRAI_COORDINATED	1.00	1.00	1.00	9209
MQTT_MALFORMED	1.00	1.00	1.00	20208
PORT_SCAN	1.00	1.00	1.00	103
SSDP_AMPLIFICATION	1.00	1.00	1.00	19670
SYN_FLOOD	1.00	1.00	1.00	13486
UDP_FLOOD	1.00	0.94	0.97	4096
accuracy			1.00	138280
macro avg	0.99	0.99	0.99	138280
weighted avg	1.00	1.00	1.00	138280

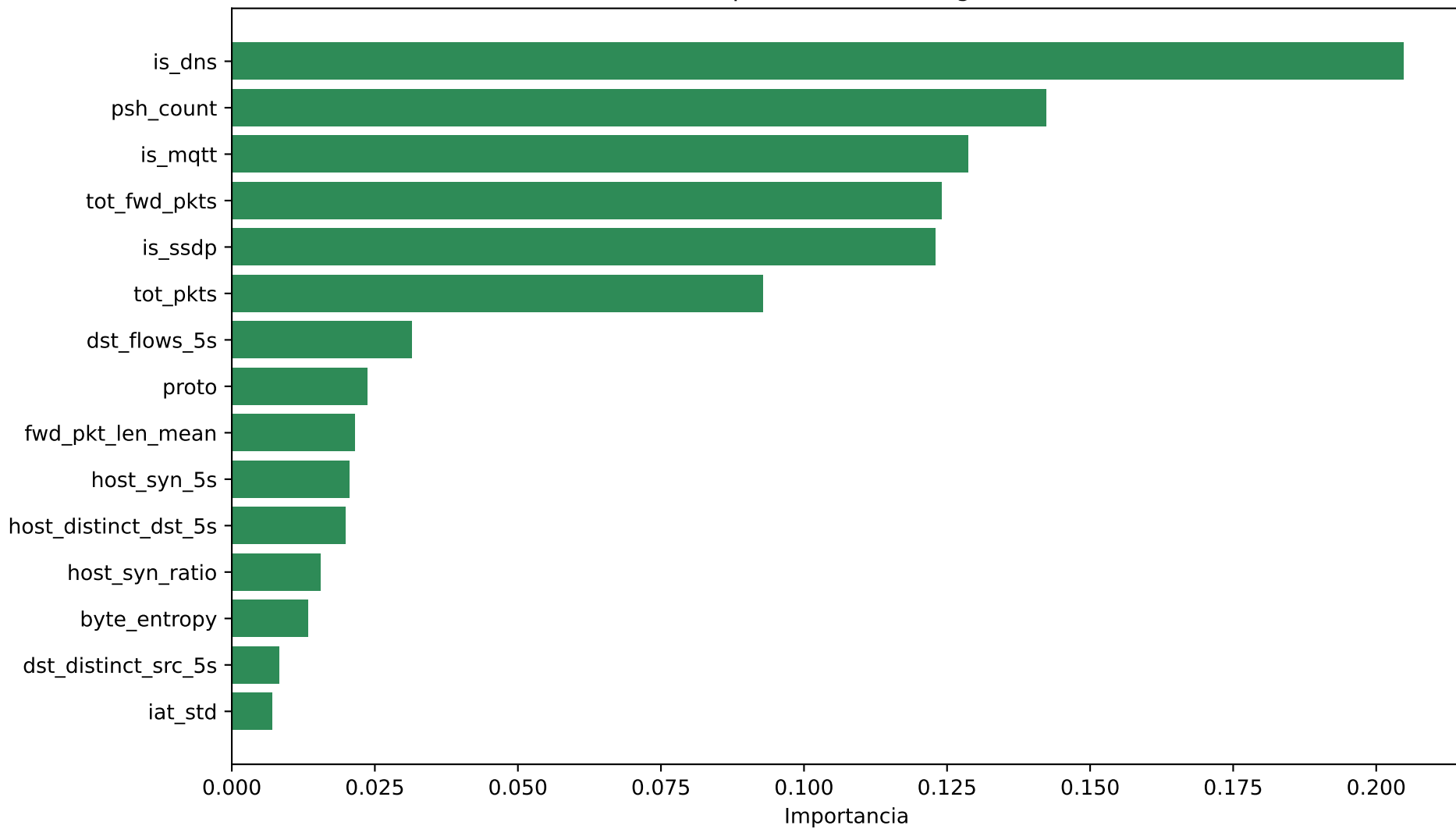
Matriz de confusión — xgb



ROC OvR — xgb



Top-15 features — xgb

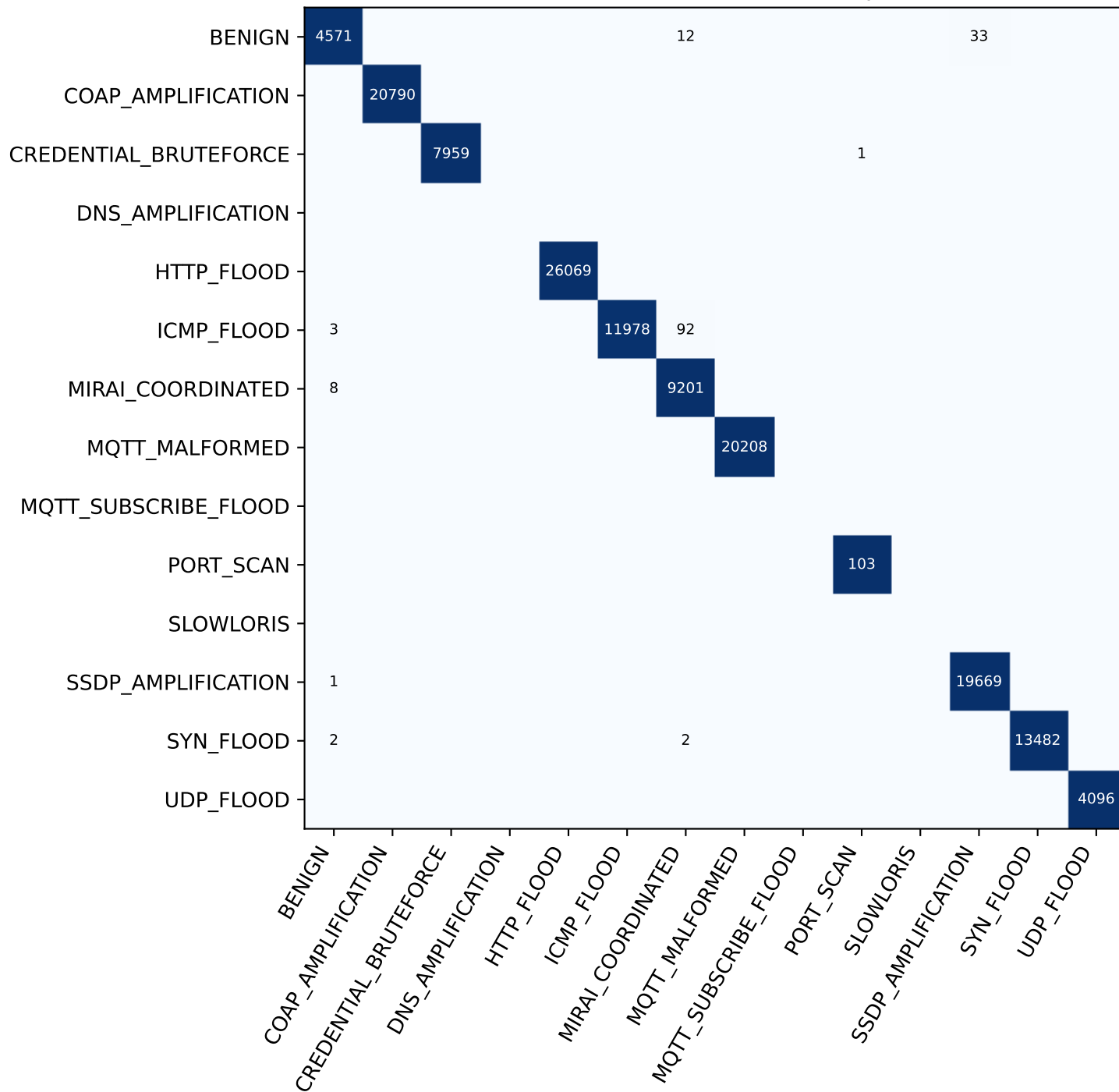


Classification report — mlp

	precision	recall	f1-score	support
BENIGN	1.00	0.99	0.99	4616
COAP_AMPLIFICATION	1.00	1.00	1.00	20790
CREDENTIAL_BRUTEFORCE	1.00	1.00	1.00	7960
HTTP_FLOOD	1.00	1.00	1.00	26069
ICMP_FLOOD	1.00	0.99	1.00	12073
MIRAI_COORDINATED	0.99	1.00	0.99	9209
MQTT_MALFORMED	1.00	1.00	1.00	20208
PORT_SCAN	0.99	1.00	1.00	103
SSDP_AMPLIFICATION	1.00	1.00	1.00	19670
SYN_FLOOD	1.00	1.00	1.00	13486
UDP_FLOOD	1.00	1.00	1.00	4096
accuracy			1.00	138280
macro avg	1.00	1.00	1.00	138280
weighted avg	1.00	1.00	1.00	138280

Matriz de confusión — mlp

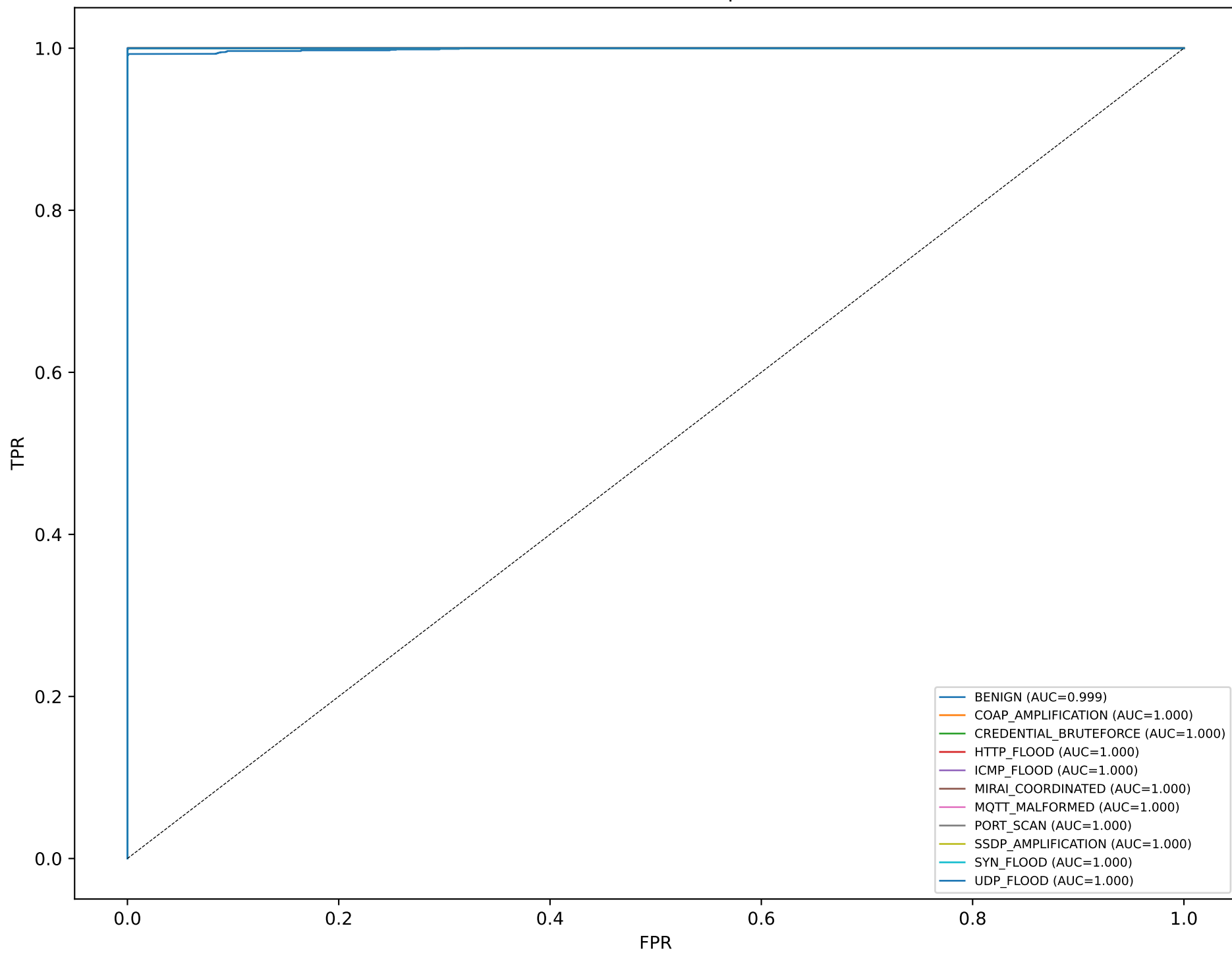
Real



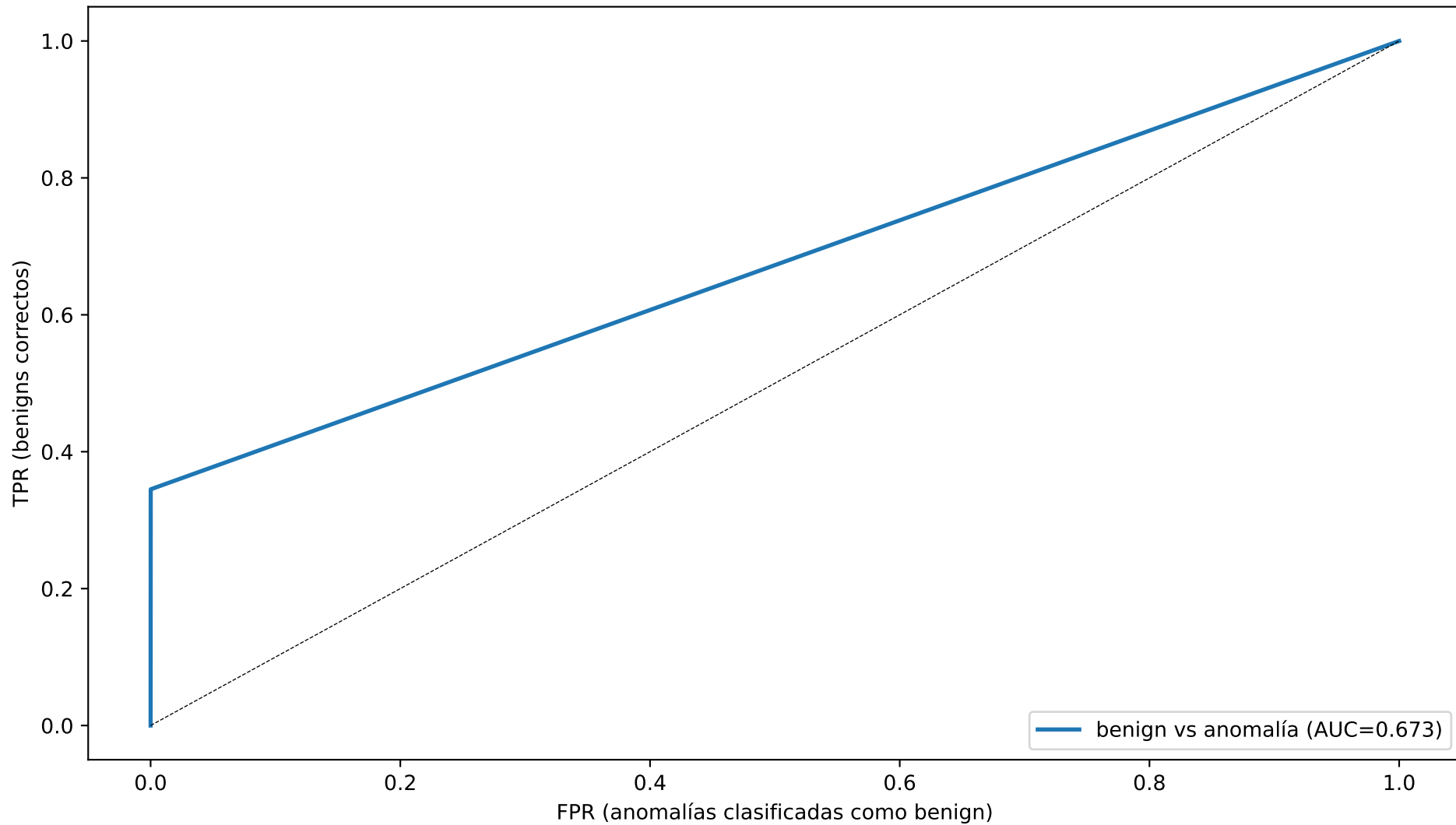
% por fila



ROC OvR — mlp



ROC binaria — isoforest



ROC binaria — autoencoder

